The transfer function of subset-sum landscapes: rigidity of the gap series off centre

(working title — scaffolding, round 88)

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Abstract

[**TODO** written last, after the theorem statements are fixed.] The three things this paper is to establish: (i) that $\mathrm{lm}_A(n)/\deg_A(n) \to \Gamma(A)$ holds for every fixed target fraction ρ, not only at the centre — so that Γ is an invariant of the landscape and not of one point of it; (ii) that the ratio is a universal functional of the *local logarithmic slope* of the representation count alone, through an explicit transfer function Φ; and (iii) that the hypothesis the first statement needs is exactly the convergence of the series that gives the second one — with a profile for which it fails.

1 Introduction

[**TODO** full introduction.] Skeleton of the argument, and what each part rests on:

- The companion papers establish $\mathrm{lm}/\deg \to \Gamma$ *at the centre* $n = \lfloor T/2 \rfloor$: paper 1 reduces it to a flatness hypothesis and paper 2 discharges that hypothesis unconditionally. The obvious question — whether Γ is a property of the landscape or of its centre — is what this paper answers.
- The combinatorial layer (classification, exact stratification, sandwich) was proved *for every target* in paper 1 and is formally verified; nothing there needs redoing. [Lean-verified]
- The minor-arc estimates of paper 2 bound $|G_B(\theta)|$ as a function of θ only. They do not refer to the extraction point m, which enters $r_B(m) = \frac{1}{2\pi}\int F_B(\theta)e^{-im\theta}d\theta$ only through a phase. They therefore transfer to off-centre targets unchanged. [quoted from paper 2]
- What is new is the middle: a tilted local limit theorem and a cancellation lemma for the odd orders in λ. [**TODO** fable-5, `spec_t3rigid_r085.md`]
- One consequence of quoting rather than reproving deserves saying at the outset: *the only ineffective constant in the trilogy is one and the same Siegel–Walfisz constant, and it enters each paper at exactly one point* — here, part (b) of Lemma 4.10. See §7.

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2 Notation, and what is inherited

[**TODO** transcribe the definitions from paper 1 in a form that does not assume $n = T/2$.] Fix throughout an increasing sequence $A = (a_1, \ldots, a_k)$ of odd positive integers, $T = \sigma(A)$, $S_2 = \sum a_i^2$, $V = S_2/4$, and a target n. Write $\rho = n/T$ and $x = \frac{1}{2} - \rho$. For $d \geq 1$ put

$$N_d = \#\{i : a_i \leq 2d\}, \qquad \sigma_d = \sum_{a_i \leq 2d} a_i, \qquad s_d = \sum_{a_i \leq 2d} a_i^2, \qquad \delta_d = d + \tfrac{\sigma_d}{2},$$

and let $B_d = \{a_i : a_i > 2d\}$ be the layer-d ensemble, of variance $V_d = (S_2 - s_d)/4 = V - s_d/4$.

Definition 2.1 (local logarithmic slope)**.** $\lambda(n)$ is the logarithmic derivative of the representation count at the target,

$$\lambda(n) \;=\; -\frac{d}{dm} \log r_A(m)\Big|_{m=n} \;=\; \tfrac{1}{4} \log \frac{r_A(n-2)}{r_A(n+2)} + O(\cdot).$$

Remark 2.2 (why the slope and not the offset). The Gaussian approximation $\lambda \approx (T/2-n)/V$ is what one writes down first, and it is wrong at the 10% level for the purposes of this paper. [**TODO** write out: the substitution turns a 15–18% drift in the measured coefficient into 0.03%, and it is the reason an earlier draft of this programme reported a spurious non-integer exponent.] This is not a cosmetic choice: every statement below is false at the stated precision if $(T/2 - n)/V$ is substituted for λ.

Definition 2.3 (the tilt)**.** P_s is the product measure on subsets under which a_i is present with probability $p_a = (1 + e^{sa})^{-1}$, and $s = s(n) > 0$ is chosen so that $\sum_a a\, p_a = n$; thus P_s is the exponential tilt of the uniform measure whose mean sits at the target.

Remark 2.4 (the tilt and the slope agree to first order). s and λ are the same quantity to leading order but not exactly: s makes the tilted *mean* equal n, whereas λ is the slope of $\log r_A$ *at* n, and the two differ by the mean–mode gap. Measured on three profiles,

$$\frac{s(n)}{\lambda(n)} \;=\; 1 + \frac{c_A}{k} + O(k^{-2}), \qquad c_A \approx 1.85 \text{ (odds)},\; 2.03 \text{ (primes)},\; 2.87 \text{ (squares)},$$

stable in x and clean in k over $k = 100, \ldots, 220$. [experimentally confirmed; `tilt_r088`] The statements below are formulated in λ, and *not* in s: substituting s multiplies the residual of Table 1 by three, exactly as the λ^2 leading behaviour of Φ predicts. [experimentally confirmed; `sres_r088`]

3 Rigidity

Theorem 3.1 (rigidity of the gap series)**.** *[proof skeleton: spec_t3rigid_r087 as amended in §4.2] Let $P = P_k$ be the first k odd primes and let $\rho \in (0,1)$ be fixed. Then*

$$\frac{\mathrm{lm}_P(\lfloor \rho T \rfloor)}{\deg_P(\lfloor \rho T \rfloor)} \;\xrightarrow[k\to\infty]{}\; \Gamma(P),$$

uniformly for ρ in compact subsets of $(0,1)$. For a general profile A the same holds under hypothesis (H) *of §6.*

$\Gamma(A) = \sum_i a_i 2^{-i}$ is therefore an invariant of the landscape and not of its centre. Theorem 3.1 is the case $\lambda \to 0$ of Theorem 4.1: at fixed ρ one has $\lambda(n) \asymp x/N \to 0$, and Φ is continuous at 0 with $\Phi(0) = \Gamma$, so

$$\Phi(\lambda) - \Gamma \;=\; O\Big(\lambda^2 \sum_{d \geq 1} 2^{1-N_d} \delta_d^2\Big) \;\longrightarrow\; 0,$$

the sum being finite precisely by (H). [**TODO** write out the uniformity in ρ; it comes from $\kappa(\rho)$ being bounded below on compacta, not from any new estimate.]

[experimentally confirmed] The evidence the theorem is modelled on. With $\mathrm{ratio}(\rho, k) = [\mathrm{lm}/\deg]/\Gamma(P_k)$, the spread over ρ contracts monotonically:

k	40	52	64	76	88	100
$\max_\rho - \min_\rho$	0.01051	0.00283	0.00257	0.00162	0.00113	0.00083
$\max_\rho \lvert\mathrm{ratio} - 1\rvert$	0.00921	0.00276	0.00244	0.00156	0.00109	0.00081

over $\rho \in [0.20, 0.50]$, by exact integer computation. [**TODO** full table; state the granularity guard $\deg \geq 10^4$ and the two correctness checks (agreement with brute force over all 2^k subsets; the reflection identity $\mathrm{lm}(n) = \mathrm{lm}(T - n)$, which is exact).]

4 The transfer function

Theorem 4.1 (transfer)**.** *[proof skeleton: spec_t3rigid_r087 as amended in §4.2] Let $A = P_k$ be the first k odd primes, or a general profile satisfying* (H)*, and let $n = \lfloor \rho T \rfloor$ with $\rho \in (0, 1)$ fixed. Then*

$$\frac{\mathrm{lm}_A(n)}{\deg_A(n)} \;=\; \Phi_k(\lambda(n)) \;+\; E_k, \qquad \Phi_k(\lambda) = 1 + \sum_{d \geq 1} 2^{1-N_d}\, e^{-\lambda^2 s_d/8} \cosh(\lambda \delta_d), \tag{1}$$

with $E_k \to 0$ as $k \to \infty$, uniformly for ρ in compact subsets of $(0, 1)$.

Remark 4.2 (what the theorem does and does not claim about E_k)*.* The assertion is $E_k \to 0$ in absolute terms, nothing sharper. The measured rate $E_k \asymp c_A/k$ of Remark 4.7 is *not* part of the statement; it is quoted as [experimentally confirmed, $k \leq 220$, three profiles] and stated as an open problem in §7. Distinguishing the two matters here more than usual, because the numerical agreement is far better than what is proved.

Three features of (1) deserve to be stated before any proof.

Remark 4.3 (the ratio sees the target only through one number)*.* Φ depends on A alone. The target enters only through $\lambda(n)$. This is a stronger statement than rigidity: rigidity says the limit does not move, (1) says how it fails to move.

Remark 4.4 ($\lambda = 0$ recovers paper 1)*.* The layer weight $e^{-\lambda^2 s_d/8}$ is 1 at $\lambda = 0$, so $\Phi(0) = 1 + \sum_{d \geq 1} 2^{1-N_d}$, which is the window identity $W_D(A) = \Gamma(A) + (2D+1)2^{-k}$ of paper 1 — already formally verified. Numerically, $\Phi(0) - \Gamma(A)$ is 0, $-4.7 \cdot 10^{-14}$ and $-2.7 \cdot 10^{-15}$ for the three sequences of Table 1.

Remark 4.5 (each layer is a smaller ensemble)*.* The factor $e^{-\lambda^2 s_d/8}$ is not a fudge: it is what remains when the Gaussian normalisations of B_d and of A fail to cancel, because $V_d = V - s_d/4$. Expanding to second order gives the coefficient of λ^2 as $\delta_d^2 - s_d/4$ rather than δ_d^2. [**TODO** write out the two lines; note that the correction is termwise smaller than the term it corrects, $s_d/4 \leq d\sigma_d/2 < \delta_d^2$, so the sum stays positive.]

4.1 Verification

[experimentally confirmed] Table 1 evaluates (1) at the measured $\lambda(n)$ against the exactly computed $\mathrm{lm}/\deg$.

$x = \frac{1}{2} - \rho$	0.06	0.08	0.10	0.15	0.20	0.25	0.30
odds, $k = 220$	0.84%	0.85%	0.85%	0.86%	0.88%	0.92%	0.97%
squares, $k = 170$	1.66%	1.67%	1.68%	1.70%	1.74%	1.80%	1.89%
primes, $k = 220$	0.91%	0.92%	0.92%	0.94%	0.96%	0.99%	1.05%

Table 1: Residual $|\text{measured} - \Phi(\lambda)|$ **as a fraction of the centred signal** $\mathrm{lm}/\deg - [\mathrm{lm}/\deg]_{n=T/2}$. Absolute residuals are 10^{-5} to 10^{-6}; quoting them as significant figures of the ratio would be meaningless, since the signal itself lives in the fifth and sixth digit.

Remark 4.6 (why the residual is quoted this way). Against the *uncentred* signal the same residuals read 2% to 300%, the large values occurring precisely where the signal is smallest. Neither number is wrong; the centred one is the one the theorem is about, because $\Phi(0) = \Gamma$ is an identity and the finite-k offset $[\mathrm{lm}/\deg]_{n=T/2} - \Gamma$ is a separate quantity of size 10^{-6}. [**TODO** keep this remark: an earlier draft of the programme reported "agreement to five or six significant figures" for a model that was 12% wrong on the signal.]

Remark 4.7 (the residual obeys a $1/k$ law). Table 1 is one column of a k-scan. Writing $R_A(k)$ for the residual in the sense of the caption, the measurement over $k = 60, \dots, 220$ gives

$$k\, R_A(k) \;\longrightarrow\; \begin{cases} 1.85\% & \text{odds } (k = 60, 100, 140, 180, 220:\;\; 2.29, 2.00, 1.91, 1.87, 1.85), \\ 2.01\% & \text{primes } (k = 100, \dots, 220:\;\; 2.38, 2.06, 2.03, 2.01), \\ 2.82\% & \text{squares } (k = 130, \dots, 190:\;\; 3.78, 2.85, 2.83, 2.82). \end{cases}$$

Two things follow. First, the residual is $O(1/k)$ and therefore vanishes, which is the form Theorem 4.1 asserts; the table's 1% is a finite-k number and not a floor. Second, the squares row is the loosest of the three because c_A is larger, not because the decay is different: squares obey the same law with a constant 1.52 times that of the odd numbers. The constants c_A coincide numerically with those of Remark 2.4 — the same mean–mode gap seen twice — but substituting s for λ does not remove the residual, so the coincidence is a shared order of magnitude and not a shared cause. [experimentally confirmed; `tilt_r088`, `sres_r088`]

What they share is a driver. The saddle-point representation gives the exact identity

$$s(n) - |\lambda(n)| \;=\; \frac{K'''(s)}{2K''(s)^2}, \qquad K'' = \sum a^2 p_a(1-p_a), \quad K''' = \sum a^3 p_a(1-p_a)(1-2p_a),$$

tested pointwise to a relative error $0.54/k$ [experimentally confirmed; `saddle_r092`], and for small tilt $K'''/(2K''^2) \approx sS_4/S_2^2$, so that $c'_A = k(s/|\lambda| - 1) \to kS_4/S_2^2$. For $a_i \asymp i^\alpha$ this is $(2\alpha+1)^2/(4\alpha+1)$. The measured c_A of this remark agrees with c'_A to within 1% on four profiles:

	α	kS_4/S_2^2	c'_A	c_A	c_A/c'_A
odd numbers	1	1.8000	1.8525	1.8674	1.008
$a_i \asymp i^{3/2}$	3/2	2.2794	2.3494	2.3318	0.993
squares	2	2.7714	2.8615	2.8377	0.992
primes	—	1.9731	2.0251	2.0253	1.000

at $k = 220$. [experimentally confirmed, four profiles; `saddle_r092`, `alpha_r092`] [**TODO** fable: $c_A = c'_A$ is still a conjecture — the driver is shared, the causation is not (substituting s makes the residual worse). State it as a conjecture with the table as its evidence.]

4.2 The bridge to paper 2

Both theorems reduce to the same five parts. **P1** is the exact combinatorial stratification, proved for every target in paper 1 [Lean-verified]; **P3** is a local limit theorem for the tilted ensemble of Definition 2.3; **P4** is the assembly, in which the odd orders in λ cancel by the reflection symmetry $\mathrm{lm}(n) = \mathrm{lm}(T-n)$ [Lean-verified]; **P5** is the error budget, which converges exactly under (H). Only **P2** is new: the minor-arc estimates of paper 2 are proved for the *uniform* measure, and P3 needs them for the tilted one.

Write $\widetilde{G}_B(\theta) = \prod_{a\in B}(1 - p_a + p_a e(a\theta))$ for the tilted generating function and $S_B(\theta) = \sum_{a\in B} e(a\theta)$ for the linear exponential sum.

Lemma 4.8 (the bridge)**.** *Put $t_a = 4p_a(1-p_a)$ and $t_{\min} = \min_{a\in B} t_a$. For every θ,*

$$|\widetilde{G}_B(\theta)|^2 = \prod_{a\in B}\Big(1 - t_a \sin^2(\pi a\theta)\Big) \le \exp\Big(-t_{\min}\sum_{a\in B}\sin^2(\pi a\theta)\Big) = \exp\Big(-\tfrac{t_{\min}}{2}(b - \mathrm{Re}\, S_B(\theta))\Big).$$

Consequently, if $\mathrm{Re}\, S_B(\theta) \le \eta\, b$ *then* $|\widetilde{G}_B(\theta)| \le \exp(-t_{\min}(1-\eta)b/4)$.

Proof. The identity $|1 - p + pe(\phi)|^2 = 1 - 4p(1-p)\sin^2(\pi\phi)$ is a computation; $t_a \ge t_{\min}$ and $1 - u \le e^{-u}$ give the middle step; and $\sum_a \sin^2(\pi a\theta) = \frac{1}{2}(b - \mathrm{Re}\, S_B(\theta))$ is the double-angle formula summed. □

Remark 4.9 (why the bound is additive, and a bound that is not available)*.* It is tempting to transfer multiplicatively, as $|\widetilde{G}_B(\theta)| \le |G_B(\theta)|^\kappa$ with G_B the untilted product, since that would carry every exponent of paper 2 across at once. **No such $\kappa > 0$ exists.** At any θ with $a\theta \in \frac{1}{2} + \mathbb{Z}$ for some $a \in B$ the untilted factor $\cos(\pi a\theta)$ vanishes, so the right-hand side is 0, while the tilted factor equals $\sqrt{1 - t_a} > 0$. Equivalently $\log(1-ty)/\log(1-y) \to 0$ as $y \to 1$. The underlying inequality $1 - ty \le (1-y)^t$ is false for every $t \in (0,1)$ — $s \mapsto (1-y)^s$ is convex and therefore lies *below* its chord $1 - sy$, with deficit exactly $-(1-t)$ at $y = 1$. Lemma 4.8 avoids the issue by never dividing by a quantity that vanishes. [refuted numerically on a 40 575-point grid and structurally; `kappa_r088`]

Lemma 4.10 (the minor-arc input)**.** *[proof skeleton; parts (a) and (c) are quotations, part (b) needs writing] Let $A = P_k$ and $N = a_k$. Then*

$$\mathrm{Re}\, S_A(\theta) \le \Big(\tfrac{1}{2} + o(1)\Big)b \qquad \textit{uniformly for } \tfrac{1}{2N} \le \theta \le 1 - \tfrac{1}{2N}.$$

(a) $q \in \{1,2\}$. Near $\theta \equiv 0$ and $\theta \equiv \frac{1}{2}$ partial summation against the prime number theorem gives $|S_A(\theta)| \le Cb/(N\|\theta - a/q\|) \le b/2$ once $\|\theta - a/q\| \ge 1/(2N)$; the excluded interval around 0 is the principal arc, and around $\frac{1}{2}$ one has $\mathrm{Re}\, S_A \approx -b$, the helpful sign. Elementary; no Siegel–Walfisz.

*(b) $3 \le q \le (\log N)^{A_0}$. Splitting into residue classes and applying Siegel–Walfisz, $S_A(a/q + \beta) = (\mu(q)/\varphi(q))W(\beta) + O(\cdot)$ with $|W| \le b$, whence $\mathrm{Re}\, S_A/b \le |\mu(q)|/\varphi(q) \le \frac{1}{2}$. Same toolset as paper 2's small-q tier. [**TODO** write out; the error term is paper 2's C_1, the one ineffective constant, and it enters here in the same way and nowhere else.]*

(c) $q > (\log N)^{A_0}$. This is the case $n = 1$ of Step 2 of prop:deepminor *in paper 2, which gives $|S_A(\theta)| \ll N(\log N)^{3 - A_0/2} + N^{4/5}(\log N)^3$, i.e. $\mathrm{Re}\, S_A/b \ll (\log N)^{4 - A_0/2} = o(1)$ for $A_0 > 8$. [quoted from paper 2, verbatim]*

Remark 4.11 (the principal arc, and that the two regimes overlap)*.* Inside $|\theta| < 1/(2N)$ Lemma 4.10 says nothing, and it does not have to: there $|a\theta| \le \frac{1}{2}$ for every $a \in A$, so $\sin^2(\pi a\theta) \ge 4a^2\theta^2$ and Lemma 4.8 gives the quadratic bound $|\widetilde{G}_A(\theta)| \le \exp(-4t_{\min}V\theta^2)$

directly. This is already $\le e^{-Cb}$ for $|\theta| \ge \theta_1 := \sqrt{Cb/(4t_{\min}V)}$, and $\theta_1 < 1/(2N)$: with $C = 0.01$ the measured $\theta_1 N$ is 0.18–0.42 across the three profiles at $x = 0.10$ and $x = 0.30$. The two regimes therefore overlap, with the crossover at $\theta = 1/(2N)$ — which is also, and not by accident, the largest θ at which the quadratic bound is valid at all. What remains, $|\theta| \le \theta_1$, is the principal arc, where P3 supplies the main term rather than a bound. [experimentally confirmed; `gap_r090`] [**TODO Seam C.** The bookkeeping of the constant C against P3's local limit theorem is the one place this could still pinch; it is assigned to the P3 write-out and carried as a named seam until then. $\theta_1 \asymp \sqrt{b/V}$ is $\sqrt{b}$ times the natural LCLT width $V^{-1/2}$, which is fine because the integrand is already e^{-Cb} at the edge, but the error term should be written out rather than asserted.]

Remark 4.12 (what part (c) does *not* need). Paper 2 must control every harmonic $n\theta$ with $n \le N_0$, and pays for it with a covering lemma and the parameter $Q_2 = (\log N)^{A_0 + A_0'}$. Here only $n = 1$ occurs, so the dichotomy in Lemma 4.10 is on the single denominator $q_1(\theta)$ and no covering lemma is needed. The quotation in (c) is therefore valid on a strictly larger set of θ than paper 2's type (ii).

Lemma 4.13 (the extremiser of the additive estimate). $\max_{q\ge 3} \dfrac{\mu(q)}{\varphi(q)} = \dfrac{1}{2}$, *and the maximum is attained at $q = 6$ and at no other q.*

Proof. $\varphi(q) = 2$ holds exactly for $q \in \{3, 4, 6\}$, where μ takes the values -1, 0, $+1$; so among these the maximum is $\mu(6)/\varphi(6) = 1/2$. Every other $q \ge 3$ has $\varphi(q) \ge 4$ and hence $\mu(q)/\varphi(q) \le 1/4 < 1/2$. □

Remark 4.14 (the same $q = 6$, twice). By Lemma 4.10(b) the worst point of the additive estimate is therefore $\theta = 1/6$, with $\operatorname{Re} S_A/b \to \frac{1}{2} = \cos 60°$. Paper 2's multiplicative estimate has its worst point at the same q: $\sup_q M(q) = M(6) = \sqrt{3}/2 = \cos 30°$. **Same extremiser, same cause, two functionals** — both say that the odd primes lie in $\pm 1 \pmod 6$ and nowhere else, read once through $\prod |\cos|$ and once through $\sum \cos$. Measured on the odd primes below 10^7, $\operatorname{Re} S_A(1/6)/b = 0.499998$ against the predicted $1/2$, and $q = 6$ is the maximiser over $2 \le q \le 16$ at each of $N = 10^5, 10^6, 10^7$. [experimentally confirmed; `eta_r090`]

Remark 4.15 (the honest form of the tilt constant). $t_{\min} = t_{\min}(\rho)$ is positive for every fixed $\rho \in (0, 1)$ and bounded below on compact subsets, which is all Lemma 4.8 needs. It is *not* governed by the approximation $sN \approx 6x$: that relation is the small-x asymptote only, and the measured ratio $sN/6x$ is 1.02–1.13 at $x = 0.10$ but 1.22–1.39 at $x = 0.30$. Measured values of $t_{\min}$: 0.91, 0.89, 0.91 at $x = 0.10$ and 0.36, 0.28, 0.34 at $x = 0.30$ for the odd numbers, squares and primes. [experimentally confirmed; `tilt_r088`, `kappa_r088`]

With Lemmas 4.8 and 4.10 the tilted generating function decays like $\exp(-t_{\min}(1-\eta)b/4)$ off the principal arc — exponentially in b, which is what P5 quotes. The tilted analogue of the $M(q)$ spectrum is lost, and is not needed: the rate enters Theorem 4.1 only through $E_k = o(1)$.

5 Universality of the rate function

[experimentally confirmed, $k = 16..56$, three reference sequences] [**TODO** full table.] With $\hat{I}_A(\rho) = -k^{-1} \log(\deg_A(\lfloor \rho T \rfloor)/2^k)$ and $D_k = \hat{I}_P - \hat{I}_{\text{ref}}$, the difference tends to 0 for each of three reference sequences: the odd numbers, the arithmetic progression $6n + 5$, and a random odd sequence of the same range. The sign is a property of the reference, not a universal fact — it is positive against the odd numbers, negative against the random sequence, and changes sign near $k = 48$ against the progression.

Remark 5.1 (the primes sit near the random sequence). At $k = 56$ the distance to a random odd sequence is 0.0016 against 0.0146 to the arithmetic progression — an order of magnitude. [**TODO** develop: this is the direct evidence for the reading that the main term is universal and the arithmetic of the sequence appears only in the correction.]

6 A profile for which the hypothesis fails

The hypothesis Theorem 3.1 needs is the convergence, with room, of the series in (1) — qualitatively, that $N_d \to \infty$ fast enough. [**TODO** state it precisely once fable-5's skeleton fixes the form.]

[experimentally confirmed] It is not vacuous. For the odd-ified cubes $a_i = 2\lfloor i^3/2 \rfloor + 1$:

profile	odds	primes	squares	cubes
N_d at $d = 10$ (any k)	10	7	4	**2**
$Q(0) = \Gamma^{-1} \sum 2^{-N_d}(\delta_d^2 - s_d/4)$	20.3	50.4	916	**381 904**

Only $\{1, 9\}$ ever lie below 20, at any k, so 2^{-N_d} never damps the growing δ_d^2 and the series turns over only near $d \approx 10^4$. At $k = 70$ the finite-size offset is still $1.75 \cdot 10^{-2}$ while the signal is $1.7 \cdot 10^{-11}$: nine orders of magnitude, so the profile is not merely slow but undecidable at any accessible size. [**TODO** present this as the counterexample that makes the hypothesis non-vacuous, not as a failure.]

6.1 Two different kinds of failure

The cubes fail *effectively* and not asymptotically, and the distinction is worth keeping. Computed directly, $\sum_d 2^{-N_d}\delta_d^2$ for the odd-ified cubes settles at $1.112 \cdot 10^7$ and does not grow with k (values at $k = 60, 100, 140, 180$: $1.112 \cdot 10^7$ from $k \approx 140$ on, against 69.5 for the odd numbers, 266.1 for $\alpha = 3/2$ and 6473 for the squares — each of them k-independent as well). [experimentally confirmed; `alpha_r092`] So (H) is not what the cubes violate; what they violate is any hope of seeing the asymptotics, the constant being four orders of magnitude larger than the odd numbers'.

A profile that fails (H) outright is available, and on the other side:

Proposition 6.1 (no profile below $\alpha = 1$)**.** *Let $a_i \asymp c\, i^\alpha$ be distinct odd integers with $\alpha < 1$. Then $c \gg k^{1-\alpha}$, hence $a_1 \to \infty$, hence $N_d = 0$ for all $d < a_1/2$ and*

$$\sum_{d \geq 1} 2^{-N_d}\delta_d^2 \;\geq\; \sum_{d < a_1/2} 2d^2 \;\asymp\; a_1^3 \;\longrightarrow\; \infty.$$

So (H) *fails for every $\alpha < 1$.*

Proof. The k values $a_1 < \cdots < a_k$ are distinct odd integers, so $a_k \geq 2k - 1$; with $a_k \asymp ck^\alpha$ this forces $c \gg k^{1-\alpha}$. For $d < a_1/2$ no a_i is $\leq 2d$, so $N_d = 0$, $\sigma_d = 0$ and $\delta_d = d$, and the displayed terms are $2d^2$. ☐

Measured on an $\alpha = 1/2$ profile, $\sum 2^{-N_d}\delta_d^2 = 4923,\ 10245,\ 16608,\ 24464$ at $k = 60, 100, 140, 180$ — growing like $k^{3/2}$, as the proposition predicts. [experimentally confirmed; `alpha_r092`] The hypothesis therefore cuts the power profiles at $\alpha = 1$ exactly, and the cubes sit inside the admissible range but beyond the reach of computation.

7 Honest scope

[**TODO** in the style of paper 1 §10 and paper 2 §10.] The classification to fill in:

- *Formally verified*: the combinatorial layer, inherited from paper 1, for every target.
- *Proved*: [**TODO** after fable-5's skeleton.] The minor-arc input is quoted from paper 2 and is m-independent; paper 2's own ineffectivity through the Siegel–Walfisz constant is inherited.
- *Experimentally confirmed, with the range stated*: rigidity (§3), the transfer function to 0.8–1.9% of the signal (Table 1), universality (§5), the profile-dependent constant $4(2\alpha+1)/(\alpha+1)$ verified on four profiles.
- *Not claimed*: the residual 0.8–1.9% of Table 1 is consistent with finite size and shrinks in k, but we do not claim it is zero.

Remark 7.1 (the single ineffective constant). Precisely: every estimate used here is effective except part (b) of Lemma 4.10, whose error term is the Siegel–Walfisz error, and paper 2's appendix identifies the same constant C_1 as the only ineffective one in that paper, entering there at one point as well. Paper 1 is unconditional and effective throughout. **Ineffective is not conditional**: no statement here depends on an unproved hypothesis; what is not available is a numerical threshold beyond which the asymptotics take hold.

Remark 7.2 (approaches that did not work). [**TODO** as in paper 2, keep the dead ends with their reasons.] At least: the tilted-measure (Γ_x) reading of off-centre targets, which the data rejected — rewriting the collapse in the exact tilt parameter makes it worse, not better; and the reading of the correction as a third-cumulant effect, which was an artefact of using the Gaussian slope.

Data availability

[**TODO** scripts and logs, as in papers 1 and 2.]

References

[1] K. Amauchi, *The gap series of an integer sequence: an arithmetic invariant governing subset-sum landscapes.* [**TODO** arXiv number]

[2] K. Amauchi, *Asymptotic flatness of subset-sum landscapes of primes: the sub-peak spectrum and the constant* $\sqrt{3}/2$. [**TODO** arXiv number]